Tianze Luo (Ph.D)

Email : tianze001@ntu.edu.sg
Homepage: https://ltz0120.github.io
Mobile : +65-80320067

Research Interests

Graph Representation Learning: Graph Neural Networks; Graph Signal Processing; Graph Generation; Graph-based Recommender Systems.
Foundation Models: Large-Language Models (LLMs) and Large MultiModal Model (LMMs); Foundation Models for Graph-Structural Data.

Education

Nanyang Technological University (NTU) — Singapore
Ph.D *in Computer Science, with Alibaba-NTU-IPP programme* — *Jan. 2020 – Jan. 2024*
Supervisor: Prof. Sinno Jialin Pan
Thesis: "Improving Representation Learning on Graph-Structural Data for Classifications, Generations and Recommendations"

Nanyang Technological University (NTU) — Singapore
Master (by research) in Computer Science (part-time) — *Aug. 2017 – Aug. 2019*
Supervisor: Prof. Ah-Hwee Tan
Thesis: "Autonomous Multi-agent Collaborative Environment Exploration"

Nanyang Technological University (NTU) — Singapore
Bachelor of Engineering in Electrical and Electronic Engineering — *Aug. 2013 – Jun. 2017*
(First Class Honours, Top 10%)

Université de Technologie de Troyes (UTT) — Troyes, France
Exchange programme — *Aug. 2015 – Jan. 2016*

Working Experience

TikTok — Singapore
Machine Learning Engineer — *Apr 2024 - Present*

- **Research on Large Multimodal Models (LMMs)**: Lead the development of **Long-Video** Large Multimodal Models (LMM) for TikTok Live to perform various tasks. We design and train the Long-Video LMM to perform vision understanding and visual reasoning on long videos. The LMM development involves large-scale dataset preparation and labelling, multi-step large-scale training, efficient online inferencing, etc.
- **Global Live Business Growth**: Train LMMs to identify desired content from live rooms. The identified contents include group live, high-quality live room highlights, live rooms content labels, etc. Especially, based on our content understanding of group lives, we have built up various live room features, onboarded thousands of new group live agencies, driving substantial revenue growth in the multimillion-dollar range.
- **Global Live Content Governance**: Train LMMs to enhance content governance capabilities under multiple violation scenarios: **cross-region accounts**, **multiple accounts**, **malicious accounts**, **vulgar performance**, etc. Enforce penalties of hundreds of thousands of dollars per month.
- **Global Live Operation Efficiency Improvement**: Build up customer service chatbot for internal operations team and agency managers. The chatbot integrates RAG and LLM, which can provide comprehensive Q&A for specific customers. Build up an AI Agent following the "creator operation" workflow, greatly enhancing the creator operators' working efficiency.

Alibaba — Singapore
Alibaba-NTU-IPP Ph.D Programme — *Jan 2020 - Mar 2024*

- **Research**: Research on graph-based recommender systems. Build up user graphs for SEA countries. Research on advanced graph neural networks to capture complex relationships between users, products, and content, resulting in enhanced user engagement and conversion rates.

- **Lazada Recommendation Systems**: Developed cross-country recommender systems for Lazada using graph-based recommendation models, to enhance the recommendation performance in the Southeast Asia market, and mitigate the data deficiency and cold start problem. Designed end-to-end pipelines that analyze large-scale user data to drive personalized recommendations.
- **Ali-Express Recommendation Systems**: Developed recommender systems and algorithms for Ali-Express with transfer learning, to share the knowledge between the homepage and the item details page, and enhance the recommendation performance on both pages.

  Developed re-ranking models for modeling mutual influence between items within and across channels, to improve the click-through rate on the homepage for Ali-Express.
- **Patents**: Applied for the patents "*An Adaptive Data Augmentation Method For Deep Graph Representation Learning*", and "*A Fast Graph Generation Method Based On A Deep Diffusion Model*".

Alibaba — Hang Zhou, China
Algorithm Engineer — *Oct 2019 - Dec 2019*

- **Ali-Express**: Developed recommender systems and algorithms for Ali-Express in Taiwan and Vietnam.

Nanyang Technological University — Singapore
Project Officer — *Aug 2017 - Oct 2019*

- **Research**: Reinforcement learning methods for multi-agent systems.
- **Multi-robot System**: Lead the project in developing real-time exploration and navigation methods for multi-robot systems.

Rakuten — Tokyo, Japan
Software Engineer Intern — *May 2016 - Jul 2016*

- **Rakuten E-money**: Developed Android SDK for Rakuten E-money App "Edy", which supports online transactions and payments.

ST-Engineering — Singapore
Software Engineer Intern — *May 2015 - Jul 2015*

- **Development**: Developed a taxi navigation Android App and tested a newly developed bus system.

Publications

Peer-reviewed Conference and Journal Paper

1. Chen, R., **Luo, T.**, Fan, Z., Zou, H., Feng, Z., Xie, G., ... & Zhang, H. (2025). Datasets and Recipes for Video Temporal Grounding via Reinforcement Learning. Accepted by EMNLP 2025 Industry track.
2. Heqing Zou*, **Tianze Luo*** (co-first author), Guiyang Xie, Victor (Xiao Jie) Zhang, Fengmao Lv, Guangcong Wang, Juanyang Chen, Zhuochen Wang, Hansheng Zhang, Huaijian Zhang. "HLV-1K: A Large-scale Hour-Long Video Benchmark for Time-Speciic Long Video Understanding". Accepted by IEEE International Conference on Multimedia & Expo (ICME). (2025)
3. Yi Ding, Joon Hei Lee, Shuailei Zhang, **Tianze Luo**, Cuntai Guan. "Decoding Human Attentive States from Spatial-temporal EEG Patches Using Transformers". Accepted by International Conference on Bioinformatics and Biomedicine (BIBM). (2025)
4. Fengmao Lv, Mengting Xiong, Junlin Fang, Lingli Zhang, **Tianze Luo**, Weichao Liang, Tianrui Li. "MSTI-Plus: Introducing Non-Sarcasm Reference Materials to Enhance Multimodal Sarcasm Target Identification". Accepted by ACM Web Conference (WWW). (2025)
5. Junlin Fang, Wenya Wang, **Tianze Luo**, Yanyong Huang, Fengmao Lv. "Progressive multimodal pivot learning: towards semantic discordance understanding as humans". In ACM International Conference on Information and Knowledge Management (CIKM-24). (2024)

6. **Tianze Luo**, Yong Liu, Sinno Jialin Pan. "Collaborative Sequential Recommendations via Multi-view GNN-Transformers". Accepted by ACM Transactions on Information Systems (ACM TOIS). (2024)
7. Zhanfeng Mo*, **Tianze Luo*** (co-first author), Sinno Jialin Pan. "Conditional Graph Generation with Graph Principal Flow Network". Accepted by ACM Web Conference (WWW). (2024)
8. **Tianze Luo**, Zhanfeng Mo, Sinno Jialin Pan. "Learning Adaptive Multiresolution Transforms via Meta-Framelet-based Graph Convolutional Network". Accepted by the International Conference on Learning Representations (ICLR). (2024)
9. Bosheng Ding, Chengwei Qin, Ruochen Zhao, **Tianze Luo**, Xinze Li, Guizhen Chen, Wenhan Xia, Junjie Hu, Anh Tuan Luu, Shafiq Joty. "Data Augmentation using LLMs: Data Perspectives, Learning Paradigms and Challenges". Accepted by ACL findings. (2024)
10. **Tianze Luo**, Zhanfeng Mo, Sinno Jialin Pan. "Fast Graph Generation via Spectral Diffusion". Accepted by IEEE Transactions on Pattern Analysis and Machine Intelligence (IEEE TPAMI) (2023).
11. **Tianze Luo**, Zhanfeng Mo, Sinno Jialin Pan. "Conditional Graph Generation with Graph Principal Flow Network". International Conference on Machine Learning (ICML-23) Workshop on Structured Probabilistic Inference & Generative Modeling. (2023)
12. **Tianze Luo**, Zichen Chen, Budhitama Subagdja, Ah-Hwee Tan. "Real-time Hierarchical Map Segmentation for Coordinating Multi-Robot Exploration". IEEE Access 11 (2022): 15680-15692.
13. Quanyu Long, **Tianze Luo**, Wenya Wang, Sinno Jialin Pan. "Domain Confused Contrastive Learning for Unsupervised Domain Adaptation". Proceedings of the North American Chapter of the Association for Computational Linguistics (NAACL-22). (2022)
14. Tianbo Li*, **Tianze Luo*** (co-first author), Yiping Ke, Sinno Jialin Pan. "Mitigating Performance Saturation in Neural Marked Point Processes: Architectures and Loss Functions." Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining. (2021)
15. **Tianze Luo**, Budhitama Subagdja, Di Wang, Ah-Hwee Tan. "Multi-agent collaborative exploration through graph-based deep reinforcement learning." 2019 IEEE International Conference on Agents (ICA-19). (2019)

Preprints & Under Review

1. Wang, W., Zou, H., **Luo, T.**, Huang, R., Zhao, Y., Wang, Z., ... & Zhang, H. (2025). Video-STR: Reinforcing MLLMs in Video Spatio-Temporal Reasoning with Relation Graph. arXiv preprint arXiv:2510.10976. (2025)
2. Heqing Zou*, **Tianze Luo*** (co-first author), Guiyang Xie, Victor (Xiao Jie) Zhang, Fengmao Lv, Guangcong Wang, Juanyang Chen, Zhuochen Wang, Hansheng Zhang, Huaijian Zhang. "HourLVU: Adaptive Hierarchical Feature Integration for Comprehensive Hour-Long Video Understanding ". arXiv preprint. (2025)
3. Heqing Zou*, **Tianze Luo*** (co-first author), Guiyang Xie, Victor (Xiao Jie) Zhang, Fengmao Lv, Guangcong Wang, Juanyang Chen, Zhuochen Wang, Hansheng Zhang, Huaijian Zhang. "From Seconds to Hours: Reviewing MultiModal Large Language Models on Comprehensive Long Video Understanding". arXiv preprint. (2025)
4. Fangkai Jiao*, Bosheng Ding*, **Tianze Luo***, Zhanfeng Mo*. "Panda LLM: Training Data and Evaluation for Open-Sourced Chinese Instruction-Following Large Language Models." arXiv preprint arXiv:2305.03025 (2023).
5. **Tianze Luo**, Qiuhao Zeng, Tianbo Li, Sinno Jialin Pan."Meta-Contrast for Graph Representation Learning". Major revision at IEEE Transactions on Pattern Analysis and Machine Intelligence (IEEE TPAMI). (2022)

6. Qiuhao Zeng*, **Tianze Luo***, Boyu Wang. “Domain-Augmented Domain Adaptation”. arXiv preprint. (2022)
7. Hao, Qi, **Tianze Luo**, and Guangda Huzhang. ”Re-ranking with constraints on diversified exposures for homepage recommender system.” arXiv preprint arXiv:2112.07621. (2021)

Honors & Awards

- **Best Paper Award**: “Multi-agent collaborative exploration through graph-based deep reinforcement learning.” 2019 IEEE International Conference on Agents (ICA-19). IEEE (2019)
- **Research Distinctions**: 2014–2015 and 2015-2016 NTU Undergraduate Research on Campus (URECA) with **distinction**.
- **Scholarship**: Senior Middle Two (SM2) Scholarship (2012-2017), Singapore Ministry of Education.

Teaching Experiences

Singapore University of Social Sciences
Associate Lecturer (Part-time) — Mar 2023 - Present
CET175 Introduction to Generative AI
MKT365 Social Media Metrics & Analytics

NTU School of Computer Science and Engineering
Teaching Assistant
CZ3005 Artificial Intelligence — Jan 2022 - May 2022
SC1015 Introduction to Data Science & Artificial Intelligence — Jan 2022 - May 2022
CZ3005 Artificial Intelligence — Aug 2020 - Dec 2020

Open Source Projects

PandaLLM (Large Language Model for Chinese) with **1,000+** stars.
https://github.com/dandelionsllm/pandallm

- **Released Base Models (Pretrain and SFT)**: Panda-7B, Panda-Instruct-7B, Panda-13B, Panda-Instruct-13B, Flan-LLaMA-7B, Panda-OpenLLaMA-7B
- **Released Models for Chat (SFT)**: Panda-LLaMA-13B-Chat, Panda-LLaMA2-13B-Chat (v2)
- **Released Models for Legal Services (Pretrain and SFT)**: Legal-Panda-13B-Chat
- **Released Models for Code Generation (Pretrain and SFT)**: Code-Panda-13B-Python
- **Released Models for Information Retrieval**: Panda-Index-large-zh, Panda-Index-large-en

Technical report: Fangkai Jiao*, Bosheng Ding*, **Tianze Luo*** (co-first author), and Zhanfeng Mo*. “Panda LLM: Training Data and Evaluation for Open-Sourced Chinese Instruction-Following Large Language Models.” arXiv preprint arXiv:2305.03025 (2023).

Professional Services

Reviewer for Journals
IEEE Transactions on Automation Science and Engineering (T-ASE)
IEEE Transactions on Neural Networks and Learning Systems (TNNLS)
IEEE Transactions on Knowledge and Data Engineering (TKDE)

Reviewer for International Conferences
International Joint Conference on Artificial Intelligence (IJCAI)
Association for the Advancement of Artificial Intelligence (AAAI)
International Conference on Machine Learning (ICML)
International Conference on Machine Learning (ICML)
International Conference on Learning Representations (ICLR)
International World Wide Web Conference (WWW)
ACM Knowledge Discovery and Data Mining (KDD)
Neural Information Processing Systems (NeurIPS)
Annual Meeting of the Association for Computational Linguistics (ACL)